Gauri Agrawal

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Open to relocation nationwide

Education

Texas A&M University — College Station, TX
B.S. in Computer Science; Minors in Mathematics, Game Design and Development — *Aug. 2023 – May 2027*
- GPA: 3.797/4.0 | Lechner Scholar | Engineering Honors

Experience

Machine Learning Researcher, Data Integration, Visualization, and Exploration Lab — College Station, TX
Texas A&M University — *May 2026 – Present*
- Built a GNN pipeline converting CFD data into graph datasets to predict fluid flow (velocity, pressure, density) for Mars atmospheric reentry, using physics-informed Mach/Reynolds features.
- **Cut 746-step autoregressive rollout error ∼1,800x (99.9%)** versus a training-budget-matched baseline via curriculum pushforward training for the GNN CFD surrogate.
- Scaled GNN training with METIS graph partitioning and parallelized preprocessing across multiple angles of attack.
- Built a two-phase MLP + Conditional VAE model generating 3-atom collision trajectories, trained on **5.4M+ trajectories across 91K+ initial conditions**.

Curriculum Developer, CSCE 312 (Computer Organization) — College Station, TX
Texas A&M University — *Jun. 2026 – Present*
- Redesigned the CSCE 312 lab curriculum, migrating the course instruction set architecture to RISC-V and developing an 8-lab sequence guiding students from digital logic gates through construction of a working CPU.
- Authored student manuals, answer keys, and grading rubrics for all labs and the team-based final project.
- Collaborated with faculty to iteratively refine lab pacing, scope, and learning objectives based on prior student performance.

Undergraduate Research Assistant, LIVE Lab — College Station, TX
Texas A&M University — *Aug. 2024 – Nov. 2025*
- Developed an AI-powered patient assistance system using Retrieval-Augmented Generation (RAG) and NLP to answer stroke rehabilitation questions with context-aware, evidence-based guidance.
- Designed retrieval and response workflows to improve relevance and reliability of AI-generated answers, contributing to key system design decisions.
- Presented research at Texas A&M Student Research Week 2025, earning **2nd Place** in Computational & Data Sciences (Undergraduate).

Teaching Assistant, Discrete Structures for Computing — College Station, TX
Texas A&M University — *Aug. 2025 – May 2026*
- Supported instruction and grading for **300+ students** covering logic, set theory, relations, functions, and proof techniques.
- Led office hours and review sessions, providing structured feedback to strengthen student problem-solving and mathematical reasoning.

Projects

PaperTrail | *React, TypeScript, FastAPI, PostgreSQL, Docker* — Jul. – Aug. 2026
- Architected a full-stack research paper management application using React, TypeScript, and FastAPI with a layered API → services → repositories → models backend structure.
- Implemented Clerk authentication, JWT verification, and per-user data isolation to secure multi-tenant access.
- Built a PDF ingestion pipeline with pdfplumber for parsing and custom sentence-aware chunking to prepare documents for retrieval-augmented generation.
- Integrated the CrossRef API for DOI-based metadata import and developed a 40+ test automated suite with PostgreSQL, Alembic migrations, and Docker Compose for local development.

Boba Bliss | *Python, Flask, PostgreSQL, React* — Oct. - Nov. 2025
- Led a 5-person team as Project Lead and Scrum Master, coordinating sprint planning, GitHub workflow, and project milestones.
- Developed a full-stack point-of-sale system supporting menu management, order processing, and role-based permissions.

Technical Skills

Languages: Python, Java, C++, C#, JavaScript, TypeScript, SQL, HTML/CSS, Haskell, x86 Assembly
Frameworks/Libraries: React, FastAPI, Flask, Tailwind CSS, SQLAlchemy, pdfplumber
Machine Learning/AI: PyTorch, Graph Neural Networks, Conditional VAEs, Retrieval-Augmented Generation, NLP
Databases/Infrastructure: PostgreSQL, Alembic, Docker, Docker Compose, AWS
Tools/Concepts: Git, REST APIs, CI/CD, Agile/Scrum, RISC-V, Computer Architecture

Honors & Awards

2nd Place, Texas A&M Student Research Week – Computational & Data Sciences, Undergraduate (2025) | Lechner Scholar | Engineering Honors | ECSM 2025 Presenter